

Heterogeneous Cognitive Transformers: From Tool Use to Cognitive Coprocessors

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Abstract

Position paper. Transformers remain the learned semantic substrate, but specialized computational engines become tightly coupled cognitive coprocessors rather than only explicitly invoked tools.

1 Motivation

Most LLM systems expose external computation through explicit tool invocation: a model or harness selects a function, serializes arguments, executes it, and serializes the result back into context. This series studies a different architecture in which the Transformer is one subsystem of a heterogeneous cognitive machine. Specialized engines can recognize computational opportunities, maintain structured state, derive consequences, and make useful results available to subsequent neural computation. Conventional tool calling remains a universal fallback.

2 Core Hypothesis

Tightly coupled specialized computation can improve reliability, efficiency, and scaling without abandoning Transformers as general learned semantic machines. The decisive evidence is not isolated benchmark accuracy but improved scaling with problem size under matched model and external-capability budgets.

3 Paper-Specific Objectives

- Define the architecture and terminology.
- Contrast explicit tool calling, planner/executor systems, neuro-symbolic systems, and opportunistic coprocessing.
- Define the coupling ladder and Papers 1–7.
- State falsifiable hypotheses for reliability, efficiency, scaling, and small-model leverage.
- Survey related work including Toolformer, AlphaGeometry, dynamic solver composition, and verifiable neuro-symbolic programming.

4 Common Architecture

The program studies progressively tighter versions of

Transformer stream $\rightarrow$ recognizer/semantic event $\rightarrow$ typed micro-IR
 $\rightarrow$ coprocessor $\rightarrow$ derived micro-state $\rightarrow$ Transformer.

A semantic interrupt is an automatically detected opportunity for specialized computation. A coprocessor is a persistent or reflex computational subsystem rather than merely an API selected through natural-language description.

5 Evaluation

Use matched checkpoints and tasks. Report final-answer accuracy and component-level correctness, generated tokens, model calls, latency, intervention frequency, false interventions, engine work, context/state size, and scaling curves. Include no-coprocessor and conventional explicit-tool baselines whenever applicable. Oracle trigger/selection controls should separate interface failures from engine capability.

6 Falsification

The proposed mechanism is not supported if gains disappear under matched compute and capability controls, if intervention errors offset reliability improvements, or if conventional explicit tool calling provides equivalent quality and cost without tighter coupling.

7 Scope Guardrails

- No implementation claims.
- Do not position symbolic systems as Transformer replacements.
- PRA/native KV are later hypotheses, not requirements.

8 Series Roadmap

- Paper 0: Heterogeneous Cognitive Transformers: From Tool Use to Cognitive Coprocessors.
- Paper 1: Reflex Computation: Automatic Calculator Assistance During Autoregressive Decoding.
- Paper 2: Heterogeneous Reflex Reasoning: Multiple Symbolic Coprocessors with Typed Micro-State.
- Paper 3: Learned Semantic Interrupts for Cognitive Coprocessors.
- Paper 4: Transactional Cognitive State: Hypotheses, Provenance, Retraction, and Backtracking.
- Paper 5: Native Coprocessor Context: From Text Reinjection to Structured and KV-Level Interfaces.

- Paper 6: Learning to Offload: Co-Adaptation Between Transformers and Cognitive Coprocessors.
- Paper 7: A Heterogeneous Cognitive Runtime for Language Models.

9 Related Work

Toolformer learns to insert API calls into language-model generation. Neuro-symbolic systems such as AlphaGeometry demonstrate complementary neural guidance and symbolic deduction. Dynamic solver-composition work studies selection among formal reasoning paradigms, while recent verifiable neuro-symbolic programming makes reasoning explicit as compositions of primitives. This series focuses specifically on automatic, persistent, decoding-coupled cognitive coprocessors and progressively learned interfaces between neural and structured computation.

References

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- [2] Trinh et al. Solving Olympiad Geometry without Human Demonstrations. Nature, 2024.
- [3] Xu et al. Adaptive LLM-Symbolic Reasoning via Dynamic Logical Solver Composition. EACL, 2026.
- [4] Bhat et al. Forethought: Verifiable Reasoning from Neurosymbolic Primitive Programming. arXiv:2607.04096, 2026.